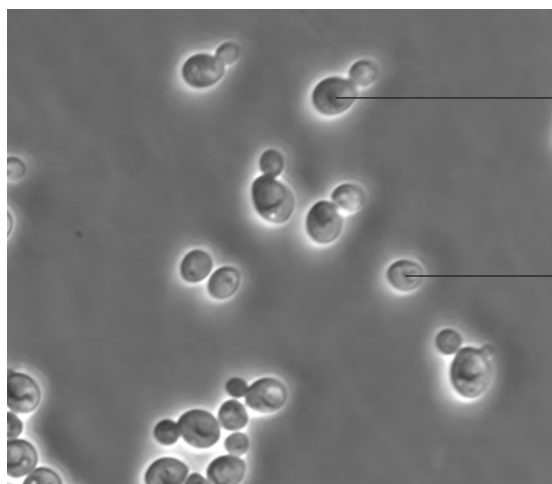


7 Yeast are single-celled eukaryotic microorganisms.

They reproduce asexually by budding. The new yeast cell (bud) grows out of the cell of an adult yeast cell following mitosis.

The photograph shows some yeast cells.



adult yeast cell with a bud

adult yeast cell without a bud

(Source: © Andre Nantel/Shutterstock)

(a) One species of yeast, *S. cerevisiae*, has a diameter from 5 μm to 10 μm .

One type of bacteria, *S. aureus*, has a diameter from 500 nm to 1500 nm.

(i) Calculate the smallest difference between the diameters of *S. cerevisiae* and *S. aureus*.

Give suitable units with your answer.

(1)

Answer

(ii) The table gives some statements about cells.

For each statement, put **one** cross ☒ in the appropriate box, in each row, to show whether these statements are true for the types of microorganisms shown.

(2)

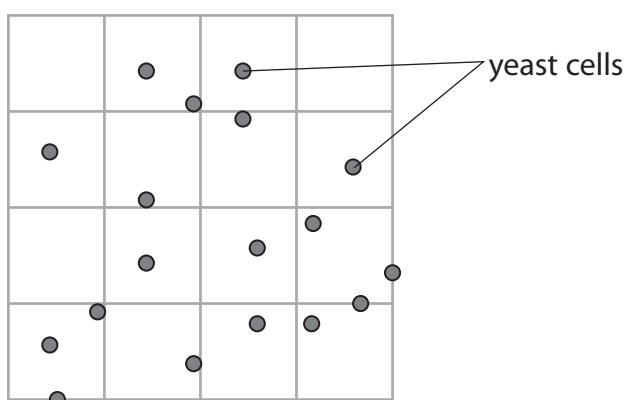
Statement about cells	Type of microorganism			
	both <i>S. cerevisiae</i> and <i>S. aureus</i>	<i>S. cerevisiae</i> only	<i>S. aureus</i> only	neither <i>S. cerevisiae</i> nor <i>S. aureus</i>
Contain both DNA and RNA	☒	☒	☒	☒
Have membranes around the cytoplasm and around the nucleus	☒	☒	☒	☒

(b) Yeast cells can be cultured and the rate of growth determined using a haemocytometer (counting chamber / Neubauer chamber).

(i) A student had a suspension of yeast cells in a liquid culture and wanted to determine the concentration of yeast cells in this culture.

A sample of this culture was added to the haemocytometer and observed under a light microscope.

The diagram shows yeast cells in part of the haemocytometer.



The volume of the sample covering this part of the haemocytometer is $0.00625 \mu\text{l}$.

$$1 \mu\text{l} = 10^{-6} \text{ dm}^3$$

Explain how to calculate the concentration of yeast cells, in cells per cm^3 , in the culture. Include in your answer an explanation of how the number of cells was determined from the diagram of a haemocytometer.

(3)

Explanation

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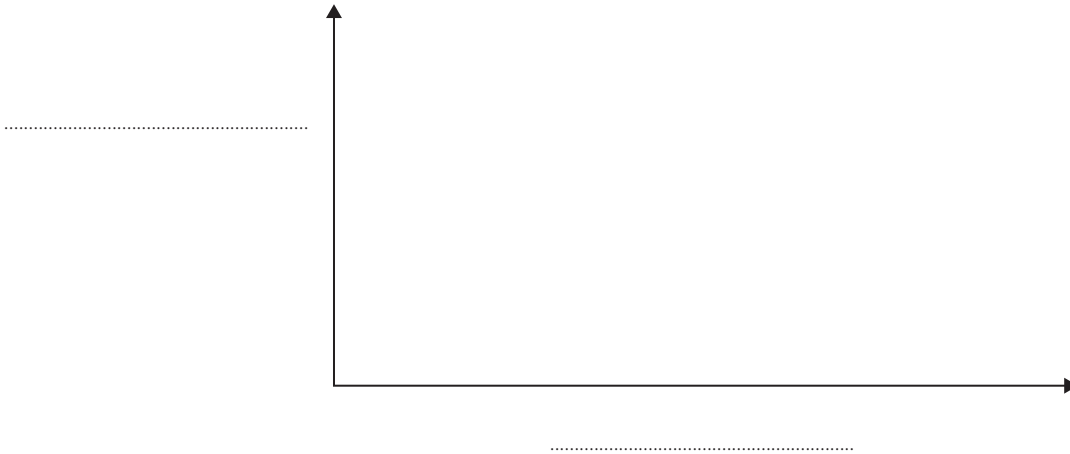
Answer cells per cm^3

- (ii) The growth curve for yeast grown in culture has four phases, similar to bacteria grown in culture.

Sketch a growth curve on the axes drawn below.

Label the axes and the four phases of the growth curve.

(4)



- (iii) Describe how the student could use a growth curve to determine the exponential growth rate constant for the yeast.

(4)

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(Total for Question 7 = 14 marks)